

Study on Factors Affecting Cognitive Health in the Elderly
Population of Shanghai

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Abstract

Based on questionnaire survey data from 1,060 elderly individuals in Shanghai, this study systematically reveals the key factors affecting the cognitive health of the elderly and their mechanisms of action. The results show that the elderly generally maintain a positive trend in self-evaluated cognitive function, but report a relatively high rate of subjective memory decline, and loneliness is prevalent, suggesting potential risks. From the dimensions of individual psychology, interpersonal interaction, community environment and technological adaptation, this study constructs a framework of the "social-ecological model" and puts forward systematic and feasible operational policies and measures, aiming to provide a theoretical basis and decision-making reference for building a "cognition-friendly" elderly health service system in Shanghai.

Keywords: Cognitive Health; Individual Psychology; Interpersonal Interaction; Community Environment; Technological Adaptation

1. Introduction

According to relevant statistics, by the end of 2025, the number of people aged 60 and above in China reached approximately 323 million, accounting for about 23% of the total population. As the first Chinese city to enter an aging society and with the highest aging level, Shanghai is home to an elderly population that already makes up 40% of its total residents, and this proportion is expected to keep rising, potentially exceeding 45% by 2050. The huge elderly population and extremely high proportion have posed greater challenges to China's response to population aging.

In addition, China's past family planning policy has resulted in smaller-sized families today. The number of empty - nest elderly households has grown rapidly. With the elderly people's declining physical conditions, their demand for daily care is rising steadily. Meanwhile, the resources available for family care are becoming increasingly scarce, leading to a sharp increase in elderly care pressure.

Against this macro background, safeguarding and promoting the cognitive health of the elderly has evolved from a purely medical issue into a major public concern related to the health of hundreds of millions of people, the well - being of millions of families, and the sustainable development of Chinese society.

Cognition is the most essential core ability that enables the elderly people to live independently, participate in social activities, and maintain mental autonomy. Cognitive decline or the progression to dementia severely undermines their quality of life and dignity. It also places heavy psychological, physical, and enormous financial

burdens on family caregivers. Furthermore, the growing number of patients will lead to a sharp increase in costs for the public health system.

Shifting the focus of cognitive health promotion and cognitive impairment prevention from passive treatment to proactive health management is a pressing task and an important path for actively responding to population aging.

2. Literature Review

Against the backdrop of population aging, cognitive health among the elderly has become a major public health and social issue. A systematic review of relevant research at home and abroad reveals that socioeconomic status, social support networks, social participation, and social psychology are all core factors influencing the cognitive health of older adults.

2.1 Demographic and Socioeconomic Factors

Age is the most important factor affecting cognitive ability, which tends to decline rapidly with advancing age. Previous studies have found that males outperform females in episodic memory, and such gender differences may be moderated by educational level, suggesting that interventions for the elderly should be more precise rather than adopting generalized measures. In addition, marital status is also an important factor influencing the cognitive function of

older adults, as it provides them with greater social support and spiritual comfort. Meanwhile, the income and wealth resources available to the elderly may affect their cognitive health in multiple ways, including access to high-quality medical care, quality of life and living conditions, and the ability to participate in various social activities.

2.2 Health and Lifestyle Factors

Relevant medical studies indicate that cognitive impairment is associated with the type, number, and management awareness of chronic diseases. Poor chronic disease management also increases the risk of cognitive impairment among the elderly, representing two major challenges in elderly health management. Mental health is closely related to cognitive function in older adults. As one of the critical risk factors for cognitive impairment, depression has a high prevalence, creating an urgent demand for mental health interventions. Data show that most older adults maintain healthy eating habits and high levels of physical exercise, both of which effectively promote cognitive health in the elderly.

2.3 Social Participation and Family Support Factors

Data from the China Health and Retirement Longitudinal Study (CHARLS) show that approximately 62.8% of adults aged 60 and above participate in one or more social activities, and the social participation rate of urban elderly is significantly higher than that of rural elderly. With changes in family living patterns, the proportion of

elderly people living alone or only with their spouses has risen rapidly, making research on the impact of different living arrangements on cognitive function highly practically significant. Furthermore, a survey has shown that about 37.9% of the elderly provide care for family members or others, among whom women bear the greatest caregiving pressure. Caregiving burden and functional limitations exert adverse effects on the mental and physical health of the elderly, which in turn directly affect their cognitive health.

Existing research has mostly focused on the impact of individual characteristics on cognitive health. My study adopts more precise quantitative analysis, focuses on the elderly population in the central urban area of Shanghai, and conducts a systematic investigation from multiple levels.

3. Methodology

3.1 Study Subjects and Sampling Methods

1. Target Population

This study takes residents aged 60 and above in Shanghai as the target population.

2. Sampling Method

The study subjects are elderly people aged 60 and above in Shanghai. Cross-sectional data were collected in the fourth quarter of 2025. Questionnaires were distributed through community neighborhood committees, senior activity centers and other channels, and a total of 1,060 valid questionnaires were finally collected.

3.2 Data Collection Methods

Structured electronic questionnaires were distributed via the professional online survey platform “Credamo”. To ensuring data quality, questionnaire completion was assisted by community staff or the authors’ schoolmates to help elderly participants who were unfamiliar with electronic devices. The survey was conducted in accordance with the principles of anonymity and voluntariness.

3.3 Data Analysis Methods

Data analysis was performed using SPSS statistical software. The specific methods include:

1. Descriptive statistics: Frequency, percentage, mean, and standard deviation analyses were conducted for the sociodemographic characteristics of the sample and the distribution of each research variable.
2. Regression analysis: Multiple linear regression was adopted as the core method. After controlling for basic variables such as age, gender, education level, and self-rated health, the independent predictive effects of variables from multiple dimensions (including social activities, social connections, digital technology use, psychological perception, and environmental factors) on the total score of “cognitive vitality” were examined respectively.
3. Reliability testing: Internal consistency reliability (Cronbach’s α) was tested for multi-item psychological scales (e.g., Cognitive Vitality

Scale, Nostalgia-Reflection Scale, Sense of Mastery Scale) to ensure the reliability of the measurement tools.

3.4 Questionnaire Item Design

The questionnaire integrated multidisciplinary assessment tools from gerontology, psychology, public health and other disciplines, covering the following modules:

1. Sociodemographic Characteristics.
2. Health Status and Lifestyle.
3. Digital Technology Use.
4. Social Participation and Support Network.
5. Community Services and Smart Elderly Care.
6. Psychological and Cognitive Assessment.

4. Results

4.1 Results of Data Analysis

This study found that the cognitive vitality of the elderly in Shanghai is influenced by an interplay of multidimensional factors.

The major positive facilitative factors include:

1. Active participation in social interaction (especially voluntary service, in-depth social contact, and physical exercise).
2. High-quality interpersonal communication (particularly in-depth conversations and written communication with family members).
3. Functional use of digital technology (e.g., proactive operations

such as mobile payment and medical registration).

4. Positive psychological capital (such as sense of control in later life and nostalgia-reflection).

5. Friendly community environment (including barrier-free facilities, activity venues, and daily convenience).

These above factors jointly maintain and promote cognitive health by enhancing the elderly's autonomy, social connection, cognitive engagement, and emotional regulation.

The main negative risk factors include:

1. Passive and low-quality social activities (e.g., only watching TV with family without communication).

2. Excessive or superficial online interaction (e.g., frequent online contact without in-depth exchange).

3. Digital anxiety and difficulties in technology use (subjective perception of difficulty in using the internet).

4. Emotional stress and negative psychological states (e.g., tension, distress, and overprotection).

5. Heavy family care burdens (e.g., stress from taking care of grandchildren).

These factors are often accompanied by the depletion of cognitive resources, social isolation, emotional burden, or reduced self-efficacy, thereby exerting adverse effects on cognitive health.

Table 1: Regression Analysis of Factors Affecting Cognitive Health

Dimension	Specific Factor	Beta	p-value	Direction	Summary
Participation in Social Interaction	Having tea, chatting and visiting with friends or neighbors	0.162	<0.001	Positive	Active and interactive social activities significantly enhance cognitive health.

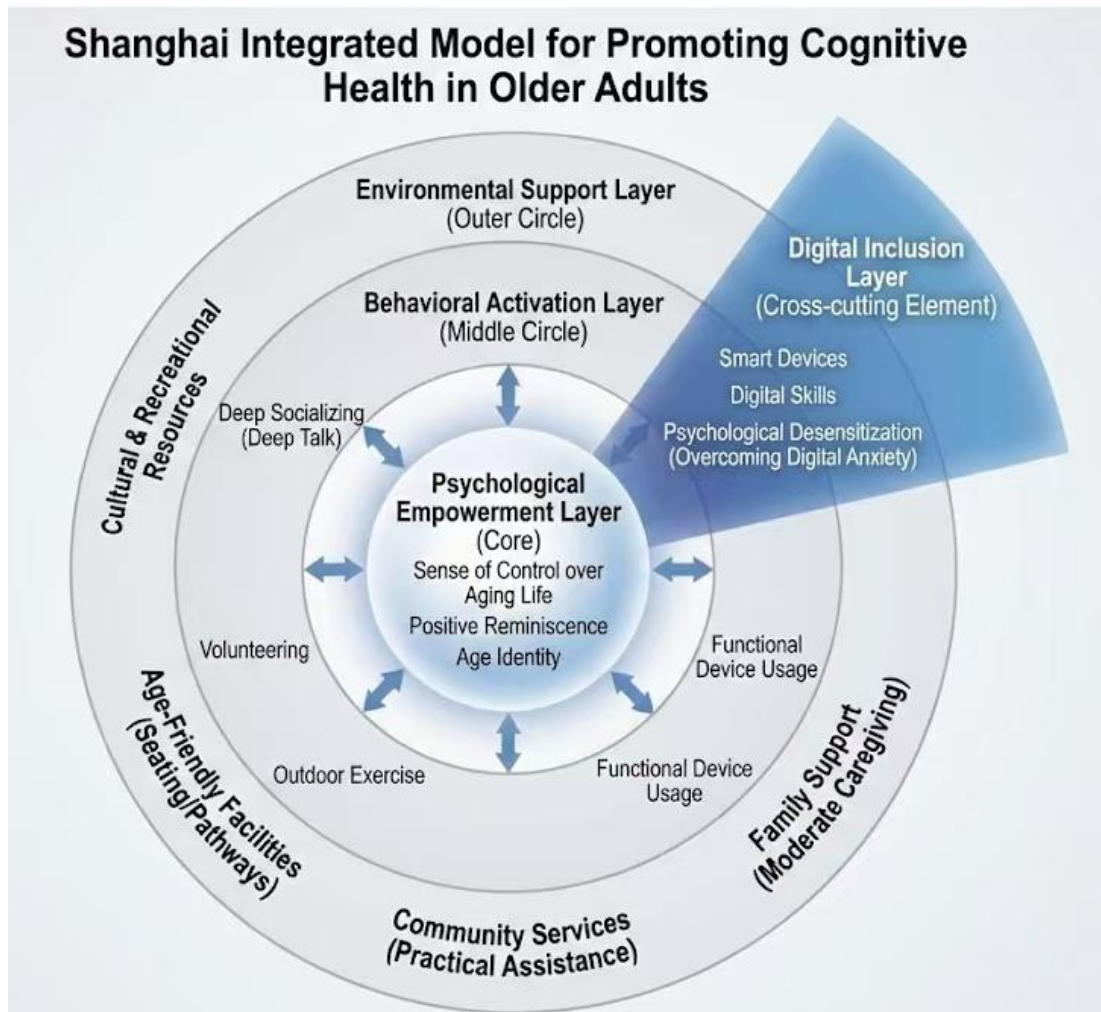
Dimension	Specific Factor	Beta	p-value	Direction	Summary
	Volunteering to help others or the community	0.156	<0.001	Positive	Volunteerism promotes self-worth and cognitive function.
	Doing physical exercise with family members or friends	0.073	0.039	Positive	Combined physical and social activities exert a synergistic cognitive benefit.
	Only watching TV with family members with little communication	-0.072	0.015	Negative	Passive entertainment lacking communication has adverse effects.
Interpersonal Communication	Online contact with children	0.097	0.006	Positive	Online contact has a positive effect on cognitive vitality.
	Online contact with friends	0.077	0.048	Positive	It alleviates loneliness but has no direct impact on cognitive vitality.
	Proportion of time spent in in-depth conversations with family members	0.101	0.001	Positive	In-depth communication is an important protective factor for cognitive vitality.
	Text-based communication (SMS / WeChat text messages)	0.091	0.030	Positive	Organized text-based communication exerts a cognitive training effect.
Use of Digital Technology	Use of smartphones	0.222	<0.001	Positive	Functional use promotes cognition.
	Use of tablet computers	0.074	0.016	Positive	
	Use of desktop / laptop computers	0.107	<0.001	Positive	
	Mobile	0.094	0.005	Positive	Practical digital

Dimension	Specific Factor	Beta	p-value	Direction	Summary
	payment and online shopping				operations enhance cognitive vitality.
	Online medical appointment and health information checking	0.066	0.041	Positive	
	Perceived ease of using the Internet	-0.294	<0.001	Negative	Digital anxiety and low self-efficacy have negative effects.
Psychological Capital	Sense of control over later life	0.403	<0.001	Positive	Autonomy and sense of control are important protective factors for cognition.
	Nostalgia and reflection	0.221	<0.001	Positive	Positive nostalgia enhances meaning in life and cognitive reserve.
	High frequency of feeling nervous or upset in the past week	-0.212	<0.001	Negative	Emotional stress significantly impairs cognition.
	Others insist on doing things for me that I am able to do myself	-0.077	0.028	Negative	Overprotection weakens self-efficacy.
Community Environment	Smooth community roads with resting seats	0.151	<0.001	Positive	Friendly physical environment promotes cognitive vitality.
	Community spaces for elderly gatherings and activities	0.131	0.002	Positive	Community social spaces benefit cognition.
	Access to a food market within 10	0.067	0.036	Positive	Living convenience is associated with

Dimension	Specific Factor	Beta	p-value	Direction	Summary
	minutes				cognitive health.
	High proportion of weekly outdoor exercise	0.150	<0.001	Positive	Outdoor activities provide multi-sensory stimulation.
	Feeling burdened by grandchild care in terms of health or time	-0.264	<0.001	Negative	Excessive caregiving burden impairs cognition.

4.2 Construction of the Integrated Model

The results of this study indicate that the cognitive health of older adults is shaped by the interaction of multiple factors, including individual psychological capital, daily life, interpersonal relationships, and the use of tools and technological environment. Therefore, an integrated model for promoting cognitive health among community-dwelling elderly people in Shanghai can be constructed based on the above dimensions.



Taking cognitive health as the core outcome, the model consists of four interacting components:

1. Psychological Empowerment Layer (Core Layer)

Psychological capital, including a sense of control over later life, positive nostalgia and reflection, and positive age identity, serves as the motivational core and protective foundation of cognitive vitality. Meanwhile, the psychological traits of self-reliance and “fear of causing trouble to others” act as a filter for accepting external elderly care services.

2. Behavioral Activation Layer (Middle Layer)

This layer includes high-quality social interactions (in-depth

communication, voluntary service), functional use of digital technology, moderate physical activity (especially outdoor exercise), and meaningful daily routines. It represents the external expression of psychological capital and directly influences cognitive functioning.

3. Environmental Support Layer (Outer Layer)

This layer comprises a friendly physical environment (barrier-free and convenient facilities), accessible and effective community services (emotional and instrumental support), a community service system conducive to behavioral activation (currently dominated by health-oriented services rather than daily and recreational services), and a family support system with reasonable intergenerational care intensity (i.e., no excessive care burden). Together, they form a supportive system for behavioral activation.

4. Digital Integration Layer (Cross-Cutting Layer)

As a cross-cutting technology and context, digital technology affects all the above dimensions. Its proper and positive use strengthens social bonds, facilitates daily life, and exerts a promoting effect on individual behavior and mental health. By contrast, the misuse of digital technology may generate more negative consequences and impose greater harm on individuals. Smart elderly care in this layer enters a “deep-water zone,” facing challenges including lack of perceived needs at the psychological level and difficulties in skill acquisition at the behavioral level.

The model emphasizes that effective interventions require multi-level coordination. For example, providing only activity spaces (environmental layer) may be insufficient. It is also necessary to design activities that promote in-depth interaction and a sense of purpose (behavioral layer), encourage older adults to perceive autonomy and choice (psychological layer), and use digital tools for promotion and organization (digital layer).

5. Discussion

Based on a comprehensive and in-depth investigation and analysis of the cognitive status of the elderly population in Shanghai, together with the above findings and the integrated model, this study provides implications for government departments and communities to formulate targeted, multi-level and implementable cognitive health intervention strategies and elderly care service optimization plans.

5.1 Psychological Empowerment and Early Identification

Carry out psychological education on active aging, popularize brain health knowledge through community health lectures, and organize group activities such as reminiscence therapy and life review. Establish a collaborative screening mechanism for cognitive health and mental health to achieve early screening and early intervention as much as possible. Promote the concept of supported decision-making, highlight “ empowerment ” rather than “replacement” in the provision of community services, and take “Supporting You to Do It Yourself” as the slogan, so that the elderly can feel a stronger sense of self-control.

5.2 Social Participation and Relationship Optimization

Promote the upgrading of community activities from

recreation-oriented to interaction-empowerment-oriented, encourage and fund community projects featuring teamwork, problem-solving, creative expression and intergenerational communication. Launch the “Family In-depth Communication Promotion Program” and organize community interaction workshops on a family basis. Build a low-cost, high-benefit mutual-aid social circle to reduce the social threshold for the elderly.

5.3 Digital Inclusion and Smart Elderly Care Support

Carry out digital skills training for the elderly, focusing on practical scenarios such as online registration, mobile payment, anti-fraud education and video interaction with grandchildren. Implement the “Three-Step Approach for Experience and Promotion of Smart Elderly Care Devices” : (1) Scenario-based experience; (2) Accompanying training; (3) Trusted recommendation. Explore the integrated model of “smart devices + services” , taking smart devices as the entry to offline services, so that the elderly can see the practical value of technology and change the perception that “they are not needed” .

5.4 Environmental Renovation and Support Network

Carry out small and beautiful age-friendly micro-renovations in the frequently used passages, stairwells and public spaces for the elderly. Optimize the supply structure of community services and

launch “life support service packages” , providing not only health services but also daily living support and cultural and recreational services for the elderly. Provide temporary care and home assistance services for the elderly undertaking heavy intergenerational care responsibilities to alleviate their long-term care pressure. In addition, formulate a clear and easy-to-understand “Community Elderly Care Service Resource Map” and establish a community elderly care resource navigation system.

6. Conclusion

Based on an in-depth survey of 1,060 elderly people in Shanghai, this study systematically examines the impacts of social participation, psychological capital, digital integration, and community environment on the cognitive health of the elderly. The findings confirm that maintaining cognitive health is a multidimensional, synergistic process. It requires not only enhancing the elderly’s internal sense of control and positive mindset, but also building a supportive external ecosystem that encourages in-depth communication, provides meaningful activities, lowers technological barriers, and reduces unnecessary burdens.

Future research should focus on translating these findings into practical and sustainable community programs and establishing a dynamic evaluation mechanism to continuously monitor intervention effects. Efforts should be made to obtain support from government and community authorities to develop one or two pilot

“cognition-friendly communities”, integrate various resources, implement comprehensive intervention experiments, and continuously refine practical models, so as to offer solutions to cognitive health challenges in the aging society of large cities.

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